

CSC15S Assignment 5 – Symmetric Encryption and File Integrate Project

Total: 100pts (due date see Blackboard)

What to turn in: zip and upload all your related files to **Assignment 5** in Blackboard. Your zip should include the following files:

- (1) CSC15Slab5.docx (with all questions completed)
- (2) log_Mission3.txt

Part 1. Base64 vs Symmetric Encryption

[20 pts. Estimated time to complete: 20 minutes]

1. Read about the Base64 Encoding using the reference below. Describe the difference between base64 and a typical symmetric encryption algorithm like AES. [10 pts]

Base64 is a binary-to-text encoding scheme that translates binary data into a sequence of printable ASCII characters, typically used for transmitting data over text-only mediums like emails or web forms. It is not a form of encryption and offers no security, as it can easily be reversed.

In contrast, AES (Advanced Encryption Standard) is a symmetric encryption algorithm that securely encrypts data by using a secret key. AES transforms data into an unreadable form and requires the correct key to decrypt it.

Reference: <https://en.wikipedia.org/wiki/Base64>

2. Finish the required reading (Chapter 4.3 of Boneh and Shoup “A Graduate Course in Cryptograph”). Describe in your own words (no more than 100), what is the linear cryptanalysis? [10 pts]

Linear cryptanalysis is a technique used to break symmetric key ciphers by finding approximate linear expressions that describe the relationship between plaintext, ciphertext, and key bits. It involves analyzing multiple known plaintext-ciphertext pairs to uncover statistical biases and correlations, which can help deduce parts of the encryption key. This method is commonly used against block ciphers like DES and requires substantial computational effort to exploit these relationships effectively.

Part 2. Dr. Evil's Training Room. Mission 3 – You've Gotta Mail

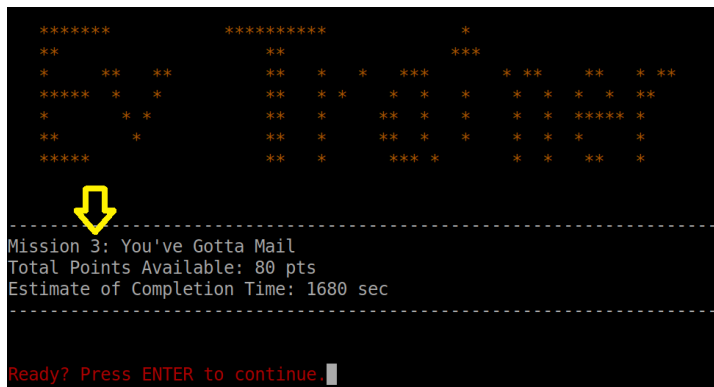
[80 pts. Estimated time to complete: 1.5 hrs]

Inspired by the 1998 movie “You’ve Got Mail”, Dr. Evil has developed an internal email exchange program named “**ygm**” for Linux users to message each other. In this mission, you will practice the use base64 encoding and AES encryption for transmitting binary files using ygm. It’s also an integrate project on testing your skills in user account management, file/dir creation and permission setting.

Please follow the instructions below to download and extract:

```
cd ~/Downloads
tar -xvf mission_ygm.tar
cd mission_ygm
chmod 755 *
sudo ./mission_ygm
```

Please make sure you are doing the right mission. See Figure below:



As usual, you need to start another terminal to complete the training. Once you feel ready to submit, please attach log_Mission3.txt to Blackboard.

The following are some additional hints for each of the challenges.

Challenge	Hint
1. Deploy	Q1 (deploy): You need to chmod on /usr/bin/ygm after you have made the copy. note the “ x ” bit for ygm is not set right. Better allow everyone in the system have the “x” access. Q2 (read mail): too easy for you, no hints given. Q3 (send email): read the manual of ygm by “ ygm --help ”.
2. Base64	Q1 (encode): the key is to have the word only. No extra white spaces! That’s the reason we use echo and file redirection. Q2 (decode): read the manual of base64 and find out what’s the extra argument that you need for decoding . Q3 (media): You need to use file redirection again, e.g., “ ygm > dump.txt ” (make sure type ENTER to finish the program). Then use VIM to chop off the contents that are not base64, in particular, make sure to CHOP OFF the trailing white spaces and new lines. Your goal is to generate the png file. So, you will need another file redirection with “ base64 -d ”.
3. AES	Q1 (decipher): note the use of passphrase and the special format required “pass:abc123”. Q2 (encrypted email): base64 decode first and then aes decipher.
4. File Permissions Again	See the <u>road map</u> below

Q4 (ygm) Suggested Roadmap

(1) create a new user, e.g., “**ygmuser**” for the ygm program and set it as the owner of **/usr/bin/ygm** and properly set up the SUID or SGUID bit (your decision to make)

(2) go to `/home/csc15s/ygm` and properly change the **owner** and **permission bits** of the **mailbox** file so that csc15s can read/write and **no others** could read/write it. But if any user runs the `/usr/bin/ygm` program, the read/write operation requested by `/usr/bin/ygm` should be allowed (so here SUID or SGID plays the trick).

*** You can test the above by switching to another user (e.g., stu101 and run the ygm command to send csc15s a message)

(3) Now **create a new user** as asked by the question (it's going to be a random user name).

*** To create a user you need **`sudo useradd uname`**

*** To reset a password of any user you need **`sudo passwd uname`**

(3.1) set up its password as requested (should be "goodyear")

(3.2) go to its home folder and create folder **ygm**

(3.3) in **ygm/** folder, create an empty text file **mailbox**

(3.4) Properly set the **owner** and **permission bits** for mailbox as well